Practice 2–3 — Relations

Solutions to Selected Problems

Problems 1.2, 2.1, 2.2, 3.3, 4.1, 7.2

Discrete Mathematics — QYSU

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1 Problem 1.2 — Domain and Range of Relations

Theory Recap: Domain and Range

For a relation $\alpha \subseteq A \times B$:

- **Domain:** $D_\alpha = \{\, x \mid \exists\, y :\ (x, y) \in \alpha \,\}$ — the set of all first coordinates.
- **Range:** $R_\alpha = \{\, y \mid \exists\, x :\ (x, y) \in \alpha \,\}$ — the set of all second coordinates.

Problem Statement

Find the domain D_α and range R_α for each of the following relations.

Method. To find the domain, fix x and ask whether **some** y exists satisfying the relation. To find the range, fix y and ask whether **some** x exists.

Intuition: Shadow Projection Analogy

Think of a relation $\alpha \subseteq \mathbb{R}^2$ as a **2-D shape** drawn on the xy-plane.

- The **domain** D_α is the *shadow* that shape casts onto the x-axis (shine a light straight down and see which x-values are "hit").
- The **range** R_α is the shadow cast onto the y-axis (shine a light from the right and see which y-values are "hit").

For example, the disk $x^2 + y^2 \leq 16$ is a filled circle of radius 4. Its shadow on each axis is the interval $[-4,\, 4]$, so both D_α and R_α equal $[-4,\, 4]$.
This "shadow" picture works for *any* geometric relation defined by an inequality or equation.

(a) $\alpha = \{(a, 1),\ (a, 2),\ (c, 1),\ (c, 2),\ (c, 4),\ (d, 5)\}$

Reading off the first coordinates and second coordinates:

$$D_\alpha = \{a,\, c,\, d\}, \qquad R_\alpha = \{1,\, 2,\, 4,\, 5\}.$$

(b) $\alpha = \{(1, 2),\ (2, 4),\ (3, 6),\ (4, 8),\ \dots\}$

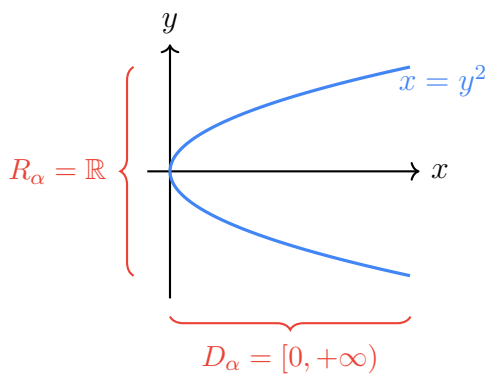
The pattern is $(n,\, 2n)$ for $n \in \mathbb{N}$, so:

$$D_\alpha = \mathbb{N} = \{1, 2, 3, \dots\}, \qquad R_\alpha = \{2, 4, 6, 8, \dots\} = 2\mathbb{N}.$$

(c) $\alpha = \{(x, y) \in \mathbb{R}^2 \mid x = y^2\}$

Since $x = y^2 \geq 0$ for all y, and for every $x \geq 0$ there exists $y = \pm\sqrt{x}$. Because squares are never negative, negative values of x cannot occur in the domain.

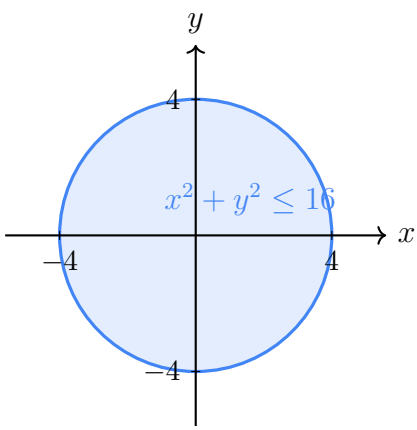
$$D_\alpha = [0, +\infty), \qquad R_\alpha = \mathbb{R}.$$



(d) $\alpha = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 16\}$

This is a closed disk of radius 4 centred at the origin. For any x with $|x| \leq 4$, there exists y satisfying $x^2 + y^2 \leq 16$, and vice versa.

$D_\alpha = [-4,\, 4], \qquad R_\alpha = [-4,\, 4].$



(e) $\alpha = \{(x, y) \in \mathbb{N}^2 \mid x \mid y\}$

Here $x \mid y$ means "x divides y." Every $x \in \mathbb{N}$ divides some y (e.g. $y = x$), so $D_\alpha = \mathbb{N}$. Every $y \in \mathbb{N}$ is divided by some x (e.g. $x = 1$), so $R_\alpha = \mathbb{N}$.

$D_\alpha = \mathbb{N}, \quad R_\alpha = \mathbb{N}.$

(f) $\alpha = \{(x, y) \in \mathbb{N}^2 \mid y \mid x\}$

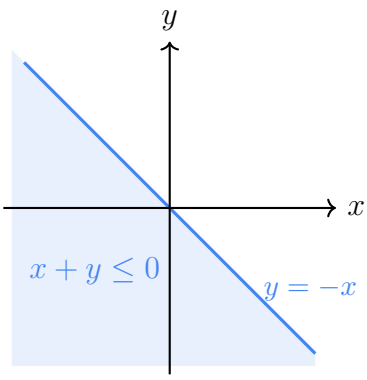
This is the inverse of part (e). By the same reasoning:

$D_\alpha = \mathbb{N}, \quad R_\alpha = \mathbb{N}.$

(g) $\alpha = \{(x, y) \in \mathbb{R}^2 \mid x + y \leq 0\}$

This is the half-plane below and including the line $y = -x$. For any $x \in \mathbb{R}$, choose $y = -x - 1$; then $x + y = -1 \leq 0$. For any $y \in \mathbb{R}$, choose $x = -y - 1$.

$D_\alpha = \mathbb{R}, \quad R_\alpha = \mathbb{R}.$



(h) $\alpha = \{(x, y) \in \mathbb{R}^2 \mid 2x \geq 3y\}$

Equivalently $y \leq \frac{2}{3}x$. For any x, choose $y = 0$ (when $x \geq 0$) or any sufficiently small y. For any y, choose $x = \frac{3}{2}y$.

$D_\alpha = \mathbb{R}, \quad R_\alpha = \mathbb{R}.$

2 Problem 2.1 — Graph and Matrix of Relations

Theory Recap: Graphs and Matrices of Relations

Let α be a relation on a finite set $A = \{a_1, \ldots, a_n\}$.

- **Graph:** Each element of A is a vertex. For a symmetric relation, if $(a_i, a_j) \in \alpha$ with $i \neq j$, draw an *undirected* edge between a_i and a_j.
- **Matrix:** The $n \times n$ matrix M with $M_{ij} = 1$ if $(a_i, a_j) \in \alpha$, and $M_{ij} = 0$ otherwise. For a symmetric relation $M = M^T$.

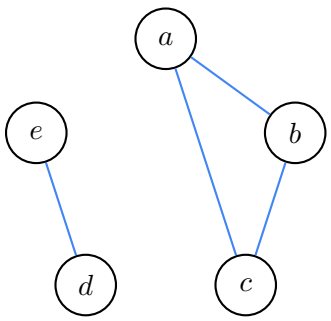
Problem Statement

Construct the graph and matrix for the following relations on $A = \{a, b, c, d, e\}$.

(a) $\alpha = \{(a,b), (b,a), (b,c), (c,b), (c,a), (a,c), (d,e), (e,d)\}$

Since every pair appears with its reverse, the relation is symmetric. Undirected edges:

$$\boxed{a\text{–}b, \quad b\text{–}c, \quad a\text{–}c, \quad d\text{–}e.}$$



Color-coded matrix (symmetry visible):

Diagonal = gray, mirrored 1's = blue / rose.

$$M_\alpha = \begin{array}{c|ccccc} & a & b & c & d & e \\ \hline a & 0 & 1 & 1 & 0 & 0 \\ b & 1 & 0 & 1 & 0 & 0 \\ c & 1 & 1 & 0 & 0 & 0 \\ d & 0 & 0 & 0 & 0 & 1 \\ e & 0 & 0 & 0 & 1 & 0 \end{array}$$

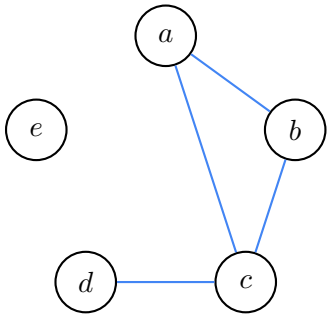
The matrix is symmetric ($M = M^T$): every shaded 1 above the diagonal has a matching shaded 1 below it. Two connected components: the triangle $\{a, b, c\}$ and the edge $\{d, e\}$.

Construction order. For a finite relation on A: (1) list the ordered pairs, (2) fill the matrix row by row from the pairs, (3) read the directed graph off the matrix.

(b) $\alpha = \{(a,b), (b,a), (b,c), (c,b), (c,d), (d,c), (c,a), (a,c)\}$

Undirected edges:

$$\boxed{a\text{–}b, \quad b\text{–}c, \quad c\text{–}d, \quad a\text{–}c.}$$



$$M_\alpha = \begin{pmatrix} & a & b & c & d & e \\ \hline a & 0 & 1 & 1 & 0 & 0 \\ b & 1 & 0 & 1 & 0 & 0 \\ c & 1 & 1 & 0 & 1 & 0 \\ d & 0 & 0 & 1 & 0 & 0 \\ e & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Triangle $\{a, b, c\}$ with d attached to c; vertex e is isolated.

(c)

Part (c) is identical to part (b) in the original problem sheet. The graph and matrix are the same as above.

3 Problem 2.2 — Digraph and Matrix of Relations

Theory Recap: Directed Graphs (Digraphs)

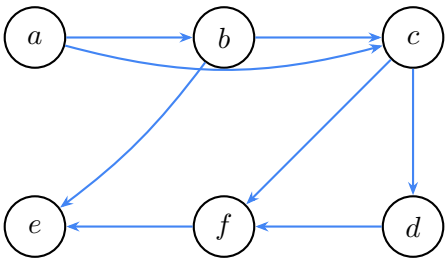
For a (not necessarily symmetric) relation α on A:

- **Digraph:** Vertices are elements of A. For each $(a_i, a_j) \in \alpha$, draw an arrow $a_i \to a_j$. A pair (a_i, a_i) is a *loop*.
- **Matrix:** Same construction; the matrix need not be symmetric.

Problem Statement

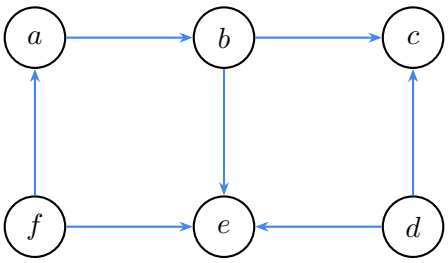
Construct the digraph and matrix for the following relations on $A = \{a, b, c, d, e, f\}$.

(a) $\alpha = \{(a,b), (b,c), (a,c), (b,e), (c,f), (c,d), (d,f), (f,e)\}$



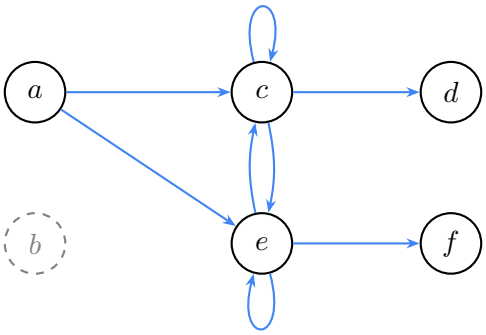
$$M_\alpha = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \\ f \end{array} \begin{array}{c} \begin{array}{cccccc} a & b & c & d & e & f \end{array} \\ \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix} \end{array}$$

(b) $\alpha = \{(a,b), (b,c), (d,c), (d,e), (f,e), (f,a), (b,e)\}$



$$M_\alpha = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \\ f \end{array} \begin{array}{c} \begin{array}{cccccc} a & b & c & d & e & f \end{array} \\ \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \end{pmatrix} \end{array}$$

(c) $\alpha = \{(a,c),(c,c),(c,d),(c,e),(e,c),(e,e),(e,f),(a,e)\}$



Note: vertex b (dashed) is completely isolated — no edges touch it.

$$M_\alpha = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \\ f \end{array} \begin{array}{c} \begin{array}{cccccc} a & b & c & d & e & f \end{array} \\ \begin{pmatrix} 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array}$$

The loops at c and e appear as 1's on the main diagonal. The b-row and b-column are all zeros.

A zero row in the matrix means "no outgoing edges" from that element. A zero column means "no incoming edges" to that element.

Contrast: Symmetric (2.1) vs. Non-symmetric (2.2) Matrices

The matrices from Problems 2.1 and 2.2 illustrate the structural difference between symmetric (undirected) and general (directed) relations. Below we place the matrix from 2.1 (a) next to the matrix from 2.2 (a):

Problem 2.1 (a) — Symmetric

$$\begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

$M = M^T$: entry (i, j) always equals entry (j, i).
Undirected edges; no arrows needed.

Problem 2.2 (a) — Non-symmetric

$$\begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

$M \neq M^T$: e.g. $M_{12} = 1$ but $M_{21} = 0$.
Directed edges (arrows) required.

Degree Tracking — Problem 2.2 (a)

For a directed relation, each vertex has an **in-degree** (number of incoming edges) and an **out-degree** (number of outgoing edges). These are read directly from the matrix: the out-degree of vertex v is the row sum, and the in-degree is the column sum.

Vertex	**In-degree**	**Out-degree**
a	0	2
b	1	2
c	2	2
d	1	1
e	2	0
f	2	1

Observe that $\sum$ in-degrees $= \sum$ out-degrees $= |\alpha| = 8$ (every edge contributes 1 to each total).

Vertex e has out-degree 0, making it a *sink* (no outgoing arrows). Vertex a has in-degree 0, making it a *source* (no incoming arrows).

4 Problem 3.3 — Composition of Relations

Theory Recap: Composition of Relations

Let $\alpha \subseteq A \times B$ and $\beta \subseteq B \times C$. The **composition** $\alpha \circ \beta$ (also written $\alpha \cdot \beta$) is the relation on $A \times C$ defined by:

$$(x, z) \in \alpha \circ \beta \iff \exists\, y \in B : \ (x, y) \in \alpha \ \wedge \ (y, z) \in \beta.$$

Read left-to-right: first go through α (from x to some y), then through β (from y to z).

Problem Statement

Let α and β be relations on $\mathbb{Z}$:

$$\alpha = \{(x,\, x+2) \mid x \in \mathbb{Z}\}, \qquad \beta = \{(x,\, x^2) \mid x \in \mathbb{Z}\}.$$

Find $\alpha \circ \beta$ and $\beta \circ \alpha$.

Convention. Relation composition is read **left-to-right** here: in $\alpha \circ \beta$, apply α first, then β. This is the most common source of errors.

Finding $\alpha \circ \beta$

We apply α first, then β:

- $(x, y) \in \alpha \implies y = x + 2$.
- $(y, z) \in \beta \implies z = y^2 = (x+2)^2$.

$$\alpha \circ \beta = \{\, (x,\ (x+2)^2) \mid x \in \mathbb{Z} \,\} = \{\, (x,\ x^2 + 4x + 4) \mid x \in \mathbb{Z} \,\}.$$

Finding $\beta \circ \alpha$

We apply β first, then α:

- $(x, y) \in \beta \implies y = x^2$.
- $(y, z) \in \alpha \implies z = y + 2 = x^2 + 2$.

$$\beta \circ \alpha = \{\, (x,\ x^2 + 2) \mid x \in \mathbb{Z} \,\}.$$

Verification and comparison

x	$\alpha \circ \beta$: $(x+2)^2$	$\beta \circ \alpha$: x^2+2
-2	0	6
0	4	2
1	9	3
3	25	11

Key observation: $\boxed{\alpha \circ \beta \neq \beta \circ \alpha}$ — composition of relations is *not commutative* in general.

$$x \xrightarrow{\alpha} x+2 \xrightarrow{\beta} (x+2)^2$$

$$\alpha \circ \beta$$

$$x \xrightarrow{\beta} x^2 \xrightarrow{\alpha} x^2+2$$

$$\beta \circ \alpha$$

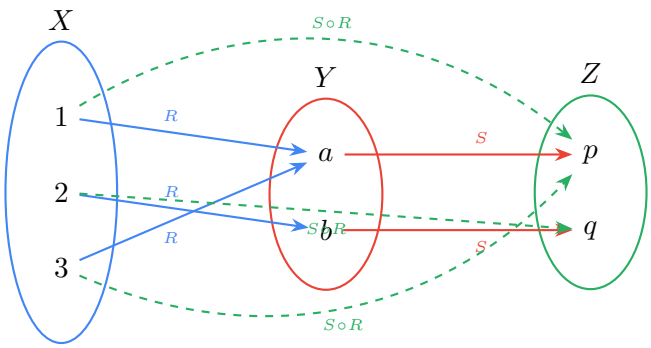
Mapping Diagram for a Finite Example

To build further intuition, consider a concrete finite example. Let $X = \{1, 2, 3\}$, $Y = \{a, b\}$, $Z = \{p, q\}$, with

$$R = \{(1, a),\ (2, b),\ (3, a)\} \subseteq X \times Y, \qquad S = \{(a, p),\ (b, q)\} \subseteq Y \times Z.$$

Then $S \circ R$ connects each $x \in X$ to every $z \in Z$ reachable via some intermediate y:

$$\boxed{S \circ R = \{(1, p),\ (2, q),\ (3, p)\}.}$$



Reading the diagram: Solid blue arrows are R, solid red arrows are S, and dashed green arrows are the composed pairs in $S \circ R$. Each dashed arrow represents a connected path through some intermediate vertex in Y.

5 Problem 4.1 — Properties of Relations

Theory Recap: Properties of Relations

A relation α on a set A may satisfy:

- **Reflexive:** $\forall x \in A: \ (x, x) \in \alpha$.
- **Symmetric:** $(x, y) \in \alpha \ \Rightarrow \ (y, x) \in \alpha$.
- **Antisymmetric:** $(x, y) \in \alpha \ \wedge \ (y, x) \in \alpha \ \Rightarrow \ x = y$.
- **Transitive:** $(x, y) \in \alpha \ \wedge \ (y, z) \in \alpha \ \Rightarrow \ (x, z) \in \alpha$.

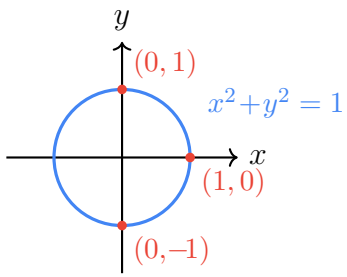
Note that symmetric and antisymmetric are *not* negations of each other — a relation can be both (e.g. Δ_A) or neither.

Problem Statement

Determine which properties each of the following relations possesses.

(a) $\alpha = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$

Geometrically, $(x, y) \in \alpha$ iff the point (x, y) lies on the unit circle.



Reflexive? No. We'd need $2x^2 = 1$ for *all* x, which fails for $x = 0$.

Symmetric? Yes. $x^2 + y^2 = y^2 + x^2$. ✓

Antisymmetric? No. $(0, 1) \in \alpha$ and $(1, 0) \in \alpha$, but $0 \neq 1$.

Transitive? No. $(1, 0)$ and $(0, 1) \in \alpha$, but $(1, 1) \notin \alpha$ since $1 + 1 = 2 \neq 1$.

Symmetric only.

Property	**Holds?**	**Justification**
Reflexive	No	$(0, 0)$: $0^2 + 0^2 = 0 \neq 1$
Symmetric	Yes	$x^2 + y^2 = y^2 + x^2$
Antisymmetric	No	$(0, 1)$ and $(1, 0)$ both $\in \alpha$, but $0 \neq 1$
Transitive	No	$(1, 0), (0, 1) \in \alpha$ but $(1, 1) \notin \alpha$
Classification	Neither equivalence relation nor partial order	

(b) $\alpha = \{(x, y) \in \mathbb{R}^2 \mid x^2 = y^2\}$

Equivalently $|x| = |y|$, so $y = \pm x$.

Reflexive? Yes. $x^2 = x^2$ always holds. ✓

Symmetric? Yes. $x^2 = y^2 \Rightarrow y^2 = x^2$. ✓

Antisymmetric? No. $1^2 = (-1)^2$ and $(-1)^2 = 1^2$, but $1 \neq -1$.

Transitive? Yes. $x^2 = y^2$ and $y^2 = z^2$ imply $x^2 = z^2$. ✓

Reflexive, symmetric, transitive — this is an **equivalence relation**.
Equivalence classes: $[x] = \{x, -x\}$ for each $x \in \mathbb{R}$.

Property	**Holds?**	**Justification**
Reflexive	Yes	$x^2 = x^2$ always
Symmetric	Yes	$x^2 = y^2 \Rightarrow y^2 = x^2$
Antisymmetric	No	$1^2 = (-1)^2$ and $(-1)^2 = 1^2$, but $1 \neq -1$
Transitive	Yes	$x^2 = y^2, y^2 = z^2 \Rightarrow x^2 = z^2$
Classification	**Equivalence relation**	

Induced partition: $\mathbb{R}$ is partitioned into blocks $\{x, -x\}$. For $x = 0$ the block is $\{0\}$; for $x \neq 0$ each block has exactly two elements.

(c) $\alpha = \{(x, y) \in \mathbb{Z}^2 \mid x \mid y\}$

Here $x \mid y$ means "x divides y," i.e. $\exists\, k \in \mathbb{Z} :\ y = kx$.

Reflexive? Yes. $x = 1 \cdot x$, so $x \mid x$ for all $x \in \mathbb{Z}$. ✓

Symmetric? No. $1 \mid 2$ but $2 \nmid 1$.

Antisymmetric? No (on $\mathbb{Z}$!). We have $2 \mid (-2)$ and $(-2) \mid 2$, but $2 \neq -2$.

Transitive? Yes. If $y = kx$ and $z = \ell y$, then $z = \ell kx$, so $x \mid z$. ✓

Reflexive and transitive, but *not* antisymmetric on $\mathbb{Z}$ (due to sign).

Property	**Holds?**	**Justification**
Reflexive	Yes	$x = 1 \cdot x$
Symmetric	No	$1 \mid 2$ but $2 \nmid 1$
Antisymmetric	No	$2 \mid (-2)$ and $(-2) \mid 2$, but $2 \neq -2$
Transitive	Yes	$y = kx,\ z = \ell y \Rightarrow z = \ell kx$
Classification	Preorder (reflexive + transitive), not a partial order	

(d) $\alpha = \{(x, y) \in \mathbb{N}^2 \mid x \mid y\}$

Same as (c) but restricted to $\mathbb{N} = \{1, 2, 3, \dots\}$ (positive integers).

Reflexive? Yes. $x = 1 \cdot x$. ✓

Symmetric? No. $1 \mid 2$ but $2 \nmid 1$.

Antisymmetric? Yes. If $x \mid y$ and $y \mid x$ with $x, y > 0$: then $y = kx$ and $x = \ell y$ with $k, \ell \geq 1$, so $x = k\ell\, x$ giving $k\ell = 1$, hence $k = \ell = 1$ and $x = y$. ✓

Transitive? Yes (same argument as (c)). ✓

Reflexive, antisymmetric, transitive — this is a **partial order**.
Compare with (c): restricting from $\mathbb{Z}$ to $\mathbb{N}$ restores antisymmetry.

Property	**Holds?**	**Justification**
Reflexive	Yes	$x = 1 \cdot x$
Symmetric	No	$1 \mid 2$ but $2 \nmid 1$
Antisymmetric	Yes	$x \mid y,\ y \mid x$ with $x, y > 0 \Rightarrow x = y$
Transitive	Yes	$y = kx,\ z = \ell y \Rightarrow z = \ell k x$
Classification	**Partial order** $(\mathbb{N}, \mid)$	

(e) $\alpha = \{(x, y) \in \mathbb{N}^2 \mid \gcd(x, y) = 1\}$

$(x, y) \in \alpha$ iff x and y are coprime.

Reflexive? No. $\gcd(2, 2) = 2 \neq 1$.

Symmetric? Yes. $\gcd(x, y) = \gcd(y, x)$. ✓

Antisymmetric? No. $\gcd(2, 3) = 1$ and $\gcd(3, 2) = 1$, but $2 \neq 3$.

Transitive? No. $\gcd(2, 3) = 1$ and $\gcd(3, 4) = 1$, but $\gcd(2, 4) = 2 \neq 1$.

Symmetric only.

Property	**Holds?**	**Justification**
Reflexive	No	$\gcd(2, 2) = 2 \neq 1$
Symmetric	Yes	$\gcd(x, y) = \gcd(y, x)$
Antisymmetric	No	$\gcd(2, 3) = 1$ and $\gcd(3, 2) = 1$, but $2 \neq 3$
Transitive	No	$\gcd(2, 3) = 1,\ \gcd(3, 4) = 1$, but $\gcd(2, 4) = 2$
Classification	Neither equivalence relation nor partial order	

(f) $\alpha = \{(x, y) \in \mathbb{Z}^2 \mid \gcd(x, y) = 1\}$

The analysis is identical to part (e) since gcd depends only on absolute values. The same counterexamples apply.

Symmetric only.

Property	**Holds?**	**Justification**
Reflexive	No	$\gcd(2, 2) = 2 \neq 1$
Symmetric	Yes	$\gcd(x, y) = \gcd(y, x)$
Antisymmetric	No	$\gcd(2, 3) = 1$ and $\gcd(3, 2) = 1$, but $2 \neq 3$
Transitive	No	$\gcd(2, 3) = 1$, $\gcd(3, 4) = 1$, but $\gcd(2, 4) = 2$
Classification	Neither equivalence relation nor partial order	

Summary Table

Relation	**Refl.**	**Symm.**	**Antisymm.**	**Trans.**	**Type**
(a) $x^2 + y^2 = 1$ on $\mathbb{R}$	—	✓	—	—	—
(b) $x^2 = y^2$ on $\mathbb{R}$	✓	✓	—	✓	equivalence
(c) $x \mid y$ on $\mathbb{Z}$	✓	—	—	✓	preorder
(d) $x \mid y$ on $\mathbb{N}$	✓	—	✓	✓	partial order
(e) $\gcd = 1$ on $\mathbb{N}$	—	✓	—	—	—
(f) $\gcd = 1$ on $\mathbb{Z}$	—	✓	—	—	—

How to check (quick tests):

- Not reflexive: find one x with $(x, x) \notin R$.
- Not symmetric: find $(a, b) \in R$ with $(b, a) \notin R$.
- Not antisymmetric: find $a \neq b$ with both (a, b) and (b, a) in R.
- Not transitive: find $(a, b), (b, c) \in R$ with $(a, c) \notin R$.

6 Problem 7.2 — Closures of a Relation

Theory Recap: Closures of a Relation

Let α be a relation on a finite set A.

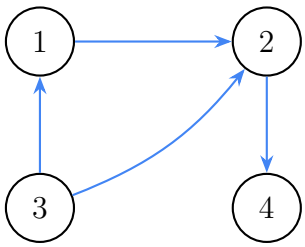
- **Reflexive closure** $r(\alpha) = \alpha \cup \Delta_A$, where $\Delta_A = \{(x, x) \mid x \in A\}$.
- **Symmetric closure** $s(\alpha) = \alpha \cup \alpha^{-1}$, where $\alpha^{-1} = \{(y, x) \mid (x, y) \in \alpha\}$.
- **Transitive closure** $t(\alpha) = \alpha \cup \alpha^2 \cup \alpha^3 \cup \cdots$ (union of all powers until no new pairs appear).

The closure adds the *fewest* pairs needed to acquire the property.

Problem Statement

Let $A = \{1, 2, 3, 4\}$ and $\alpha = \{(1, 2), (3, 1), (3, 2), (2, 4)\}$.
Find $r(\alpha)$, $s(\alpha)$, and $t(\alpha)$.

The original relation



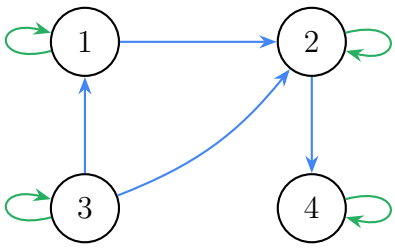
$$M_\alpha = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Rows/columns: $1, 2, 3, 4$.

Reflexive closure $r(\alpha)$

Add all diagonal pairs (x, x) for $x \in A$:

$$\begin{aligned} r(\alpha) &= \alpha \cup \{(1, 1), (2, 2), (3, 3), (4, 4)\} \\ &= \boxed{\{(1, 1), (1, 2), (2, 2), (2, 4), (3, 1), (3, 2), (3, 3), (4, 4)\}.} \end{aligned}$$



$$M_{r(\alpha)} = \begin{pmatrix} \mathbf{1} & 1 & 0 & 0 \\ 0 & \mathbf{1} & 0 & 1 \\ 1 & 1 & \mathbf{1} & 0 \\ 0 & 0 & 0 & \mathbf{1} \end{pmatrix}$$

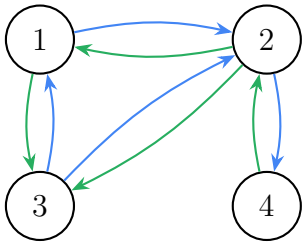
Bold = added 1's on diagonal.

Symmetric closure $s(\alpha)$

Add the reverse of every pair:

$$\alpha^{-1} = \{(2, 1), (1, 3), (2, 3), (4, 2)\}.$$

$$\boxed{s(\alpha) = \alpha \cup \alpha^{-1} = \{(1, 2), (1, 3), (2, 1), (2, 3), (2, 4), (3, 1), (3, 2), (4, 2)\}.}$$



$$M_{s(\alpha)} = \begin{pmatrix} 0 & 1 & \mathbf{1} & 0 \\ \mathbf{1} & 0 & \mathbf{1} & 1 \\ 1 & 1 & 0 & 0 \\ 0 & \mathbf{1} & 0 & 0 \end{pmatrix}$$

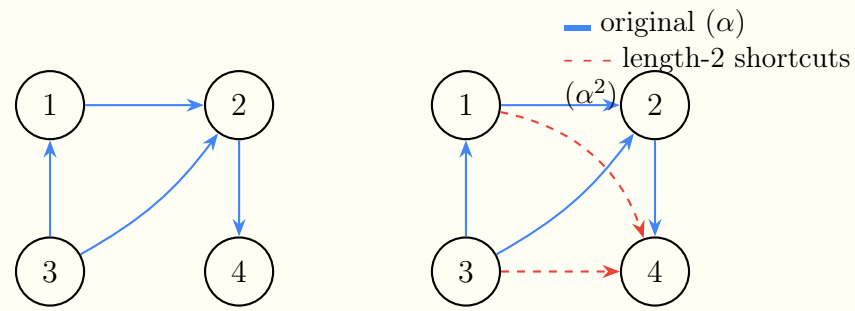
Bold = added 1's (M is now symmetric).

Transitive closure $t(\alpha)$

Interpretation. The transitive closure $t(\alpha)$ adds all pairs (a, c) such that there is a directed path from a to c in the relation graph. Equivalently, $(a, c) \in t(\alpha)$ iff a can reach c in one or more steps.

Visual Build-Up: Adding "Shortcut" Edges

The transitive closure is built layer by layer. At each step we look for paths of increasing length and add a direct "shortcut" edge for each one.



Step 0: α (original)

Step 1: add α^2 shortcuts

In this example, vertex 4 is a sink (no outgoing edges), so there are *no* paths of length 3 that produce new pairs. The process terminates: $t(\alpha) = \alpha \cup \alpha^2$.
In general, for $|A| = n$ the process always terminates after at most $n - 1$ steps, since any path in an n-vertex graph that does not repeat vertices has length $\leq n - 1$.

We compute successive powers of α:

Step 1: $\alpha^2 = \alpha \circ \alpha$. For each chain $x \xrightarrow{\alpha} y \xrightarrow{\alpha} z$:

- $(1, 2)$ and $(2, 4)$: gives $\mathbf{(1, 4)}$. **New!**

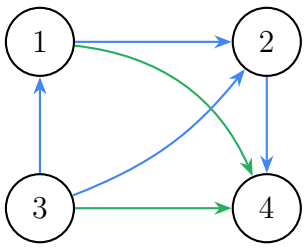
- $(3,1)$ and $(1,2)$: gives $(3,2)$. Already in α.
- $(3,2)$ and $(2,4)$: gives $\mathbf{(3,4)}$. **New!**

New pairs: $\boxed{(1,4) \text{ and } (3,4)}$.

Step 2: $\alpha^3 = \alpha^2 \circ \alpha$**.** The new second coordinates are all 4, but 4 has no outgoing edges in α, so no further pairs arise.

$$t(\alpha) = \alpha \cup \alpha^2 = \{(1,2),\ (1,4),\ (2,4),\ (3,1),\ (3,2),\ (3,4)\}.$$

Interpretation: $(x,z) \in t(\alpha)$ iff there is a directed path from x to z in the original digraph.



$$M_{t(\alpha)} = \begin{pmatrix} 0 & 1 & 0 & \mathbf{1} \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & \mathbf{1} \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Bold = added 1's.

Reachability summary:

- From 1: can reach 2 and 4 (path: $1 \to 2 \to 4$).
- From 2: can reach 4.
- From 3: can reach 1, 2, and 4 (path: $3 \to 1 \to 2 \to 4$).
- From 4: no outgoing edges — 4 is a *sink*.